

Foundations of Learning rate neural network

[Learning rate neural network]

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

1.0 Content Block

Hardware Since the 2010s, advances in both machine learning algorithms and computer hardware have led to more efficient methods for training deep neural networks that contain many layers of non-linear hidden units and a very large output layer. By 2019, graphics processing units (GPUs), often with AI-specific enhancements, had displaced CPUs as the dominant method for training large-scale commercial cloud AI. OpenAI estimated the hardware computation used in the largest deep learning projects from AlexNet (2012) to AlphaZero (2017) and found a 300,000-fold increase in the amount of computation required, with a doubling-time trendline of 3.4 months. Special electronic circuits called deep learning processors were designed to speed up deep learning algorithms. Deep learning processors include neural processing units (NPUs) in Huawei cellphones and cloud computing servers such as tensor processing units (TPU) in the Google Cloud Platform. Cerebras Systems has also built a dedicated system to handle large deep learning models, the CS-2, based on the largest processor in the industry, the second-generation Wafer Scale Engine (WSE-2). Atomically thin semiconductors are considered promising for energy-efficient deep learning hardware where the same basic device structure is used for both logic operations and data storage. In 2020, Marega et al. published experiments with a large-area active channel material for developing logic-in-memory devices and circuits based on floating-gate field-effect transistors (FGFETs). In 2021, J. Feldmann et al. proposed an integrated photonic hardware accelerator for parallel convolutional processing. The authors identify two key advantages of integrated photonics over its electronic counterparts: (1) massively parallel data transfer through wavelength division multiplexing in conjunction with frequency combs, and (2) extremely high data modulation speeds. Their system can execute trillions of multiply-accumulate operations per second, indicating the potential of integrated photonics in data-heavy AI applications.

2.0 Content Block

A common evaluation set for image classification is the MNIST database data set. MNIST is composed of handwritten digits and includes 60,000 training examples and 10,000 test examples. As with TIMIT, its small size lets users test multiple configurations. A comprehensive list of results on this set is available. Deep learning-based image recognition has become "superhuman", producing more accurate results than human contestants. This first occurred in 2011 in recognition of traffic signs, and in 2014, with recognition of human faces. Deep learning-trained vehicles now interpret 360° camera views. Another example is Facial Dysmorphology Novel Analysis (FDNA) used to analyze cases of human malformation connected to a large database of genetic syndromes.

3.0 Content Block

A main criticism concerns the lack of theory surrounding some methods. Learning in the most common deep architectures is implemented using well-understood gradient descent. However, the theory surrounding other algorithms, such as contrastive divergence is less clear. (e.g., Does it converge? If so, how fast? What is it approximating?) Deep learning methods are often looked at as a black box, with most confirmations done empirically, rather than theoretically. In further reference to the idea that artistic sensitivity might be inherent in relatively low levels of the cognitive hierarchy, a published series of graphic

representations of the internal states of deep (20-30 layers) neural networks attempting to discern within essentially random data the images on which they were trained demonstrate a visual appeal: the original research notice received well over 1,000 comments, and was the subject of what was for a time the most frequently accessed article on The Guardian's website. With the support of Innovation Diffusion Theory (IDT), a study analyzed the diffusion of Deep Learning in BRICS and OECD countries using data from Google Trends.

4.0 Content Block

Scale-up/out and accelerated DNN training and decoding Sequence discriminative training Feature processing by deep models with solid understanding of the underlying mechanisms Adaptation of DNNs and related deep models Multi-task and transfer learning by DNNs and related deep models CNNs and how to design them to best exploit domain knowledge of speech RNN and its rich LSTM variants Other types of deep models including tensor-based models and integrated deep generative/discriminative models. More recent speech recognition models use Transformers or Temporal Convolution Networks with significant success and widespread applications. All major commercial speech recognition systems (e.g., Microsoft Cortana, Xbox, Skype Translator, Amazon Alexa, Google Now, Apple Siri, Baidu and iFlyTek voice search, and a range of Nuance speech products, etc.) are based on deep learning.

5.0 Content Block

Commercial activity Facebook's AI lab performs tasks such as automatically tagging uploaded pictures with the names of the people in them. Google's DeepMind Technologies developed a system capable of learning how to play Atari video games using only pixels as data input. In 2015 they demonstrated their AlphaGo system, which learned the game of Go well enough to beat a professional Go player. Google Translate uses a neural network to translate between more than 100 languages. In 2017, Covariant.ai was launched, which focuses on integrating deep learning into factories. As of 2008, researchers at The University of Texas at Austin (UT) developed a machine learning framework called Training an Agent Manually via Evaluative Reinforcement, or TAMER, which proposed new methods for robots or computer programs to learn how to perform tasks by interacting with a human instructor. First developed as TAMER, a new algorithm called Deep TAMER was later introduced in 2018 during a collaboration between U.S. Army Research Laboratory (ARL) and UT researchers. Deep TAMER used deep learning to provide a robot with the ability to learn new tasks through observation. Using Deep TAMER, a robot learned a task with a human trainer, watching video streams or observing a human perform a task in-person. The robot later practiced the task with the help of some coaching from the trainer, who provided feedback such as "good job" and "bad job".

6.0 Content Block

Image restoration Deep learning has been successfully applied to inverse problems such as denoising, super-resolution, inpainting, and film colorization. These applications include learning methods such as "Shrinkage Fields for Effective Image Restoration" which trains on an image dataset, and Deep Image Prior, which trains on the image that needs restoration.

7.0 Content Block

Cyber threat As deep learning moves from the lab into the world, research and experience show that artificial neural networks are vulnerable to hacks and deception. By identifying patterns that these systems use to function, attackers can modify inputs to ANNs in such a way that the ANN finds a match that human observers would not recognize. For example, an attacker can make subtle changes to an image such that the ANN finds a match even though the image looks to a human nothing like the search target. Such manipulation is termed an "adversarial attack". In 2016 researchers used one ANN to doctor images in trial and error fashion, identify another's focal points, and thereby generate images that deceived it. The modified images looked no different to human eyes. Another group showed that printouts of doctored

images then photographed successfully tricked an image classification system. One defense is reverse image search, in which a possible fake image is submitted to a site such as TinEye that can then find other instances of it. A refinement is to search using only parts of the image, to identify images from which that piece may have been taken. Another group showed that certain psychedelic spectacles could fool a facial recognition system into thinking ordinary people were celebrities, potentially allowing one person to impersonate another. In 2017 researchers added stickers to stop signs and caused an ANN to misclassify them. ANNs can however be further trained to detect attempts at deception, potentially leading attackers and defenders into an arms race similar to the kind that already defines the malware defense industry. ANNs have been trained to defeat ANN-based anti-malware software by repeatedly attacking a defense with malware that was continually altered by a genetic algorithm until it tricked the anti-malware while retaining its ability to damage the target. In 2016, another group demonstrated that certain sounds could make the Google Now voice command system open a particular web address, and hypothesized that this could "serve as a stepping stone for further attacks (e.g., opening a web page hosting drive-by malware)". In "data poisoning", false data is continually smuggled into a machine learning system's training set to prevent it from achieving mastery.